Linear Regression Analysis of Ames Housing Dataset

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Find this document on my website: https://kdhenderson.github.io/projects/. The original SAS and R analysis is here.

Introduction

In this analysis, I aim to address two questions regarding home sale prices in Ames, Iowa, using various regression techniques. The first question focuses on understanding the relationship between the sale price and the square footage of the living area in three specific neighborhoods. This estimate is intended to provide valuable market-specific insights for Century 21 Ames. The second question seeks to build a predictive model for home sale prices across all of Ames, identifying which combination of home characteristics produces the most robust model and best predictions.

Data Description

The data used in these analyses are from the Ames Housing Dataset (De Cock, 2011). This dataset was obtained from the "House Prices - Advanced Regression Techniques" competition on Kaggle (Montoya, 2016). The training set has 1460 observations with 81 variables, and the testing set has 1459 observations with 80 variables, missing the 'SalePrice' response variable.

Variables of importance for these analyses:

- SalePrice: The property's sale price in dollars, the target variable.
- BsmtFinSF1: Type 1 finished square feet.
- GarageCars: Size of garage in car capacity.
- GrLivArea: Above-ground living area in square feet.
- LotArea: Lot size in square feet.
- MSSubClass: The building class.
- Neighborhood: Physical locations within Ames city limits.
- OverallCond: Overall condition rating.
- OverallQual: Overall material and finish quality.
- YearBuilt: Original construction date.

Analysis Question 1

Problem

I aim to use linear regression to estimate the relationship between home sale price and the living area in square feet within the North Ames, Edwards, and Brookside neighborhoods. I would also like to determine if this relationship depends on the neighborhood.

Build and Fit the Model

I will start by examining the simplest model,

$$\mu\{\text{Price} \mid \text{Area}\} = \beta_0 + \beta_1 \text{Area}.$$

I will also include neighborhood as an indicator variable,

$$\mu\{\text{Price} \mid \text{Area, Neighborhood}\} = \beta_0 + \beta_1 \text{Area} + \beta_2 \text{Brookside} + \beta_3 \text{Edwards},$$

and examine interaction terms with Area and Neighborhood,

$$\begin{aligned}\mu\{\text{Price} \mid \text{Area, Neighborhood}\} = \beta_0 + \beta_1 \text{Area} + \beta_2 \text{Brookside} + \beta_3 \text{Edwards} \\ + \beta_4 \text{Area} \times \text{Brookside} + \beta_5 \text{Area} \times \text{Edwards}.\end{aligned}$$

Additionally, I will explore whether any transformations of the data may be appropriate.

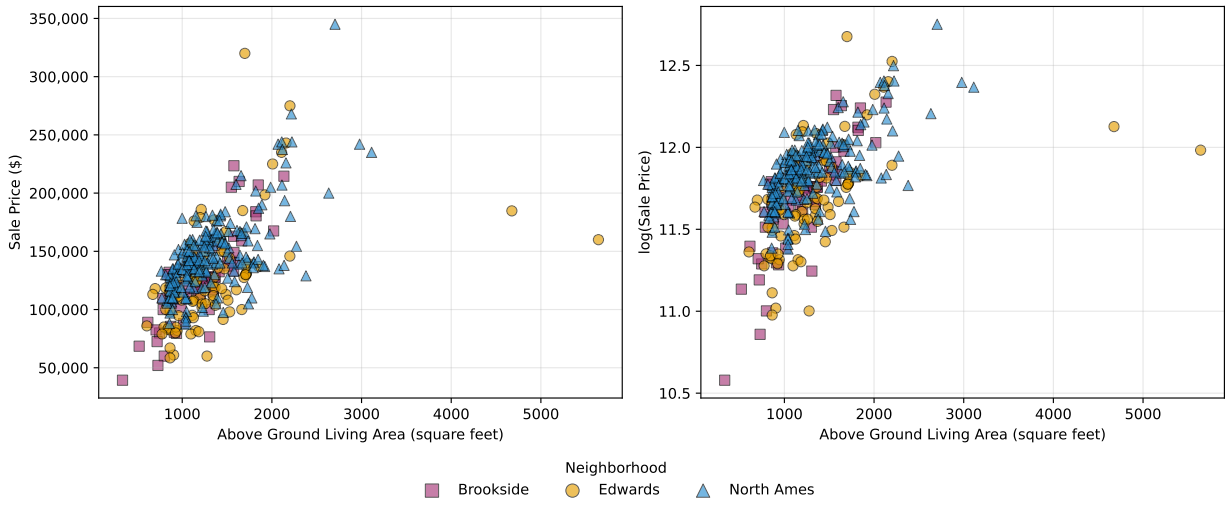


Figure 1: Scatterplot of sale price in dollars and above-ground living area in square feet with neighborhoods plotted by color and shape (pink-purple squares: Brookside, orange circles: Edwards, blue triangles: North Ames). Original, untransformed data are on the left, and data with log transformation of sale price are on the right.

After comparing the untransformed data with the log-transformed sale price as in Figure 1, the log transformation seems to provide a better linear fit. I will proceed with two candidate models: one with log-transformed price and untransformed area (log-lin), and the other with log transformations of both price and area (log-log). A comparison of these scatterplots is in Supplementary Figure 1 in the appendix.

Checking Assumptions

Residual Plots

The residual plots of the log-lin transformed data (Figure 2) satisfy the assumptions of linear regression better than those of the untransformed data. The residuals of the log-log transformed data in Supplementary Figure 2 also look quite good.

Influential point analysis (Cook's D and Leverage)

Observation 339 has a very large Cook's D, and observation 131 is also notable (Figure 3). Both houses have large living areas but low sale prices, likely due to their partial sale condition. I found four partially completed houses within the neighborhoods of interest, with at least two being influential points. Those two are also the only houses with living areas greater than 3500 square feet. Although I could restrict the data range and exclude partially completed houses above 3500 square feet, I decided to exclude all partially completed houses from this analysis, because there are only four. If analyzing all neighborhoods, this strategy might not be appropriate. In the full dataset, there are many more houses listed in partial condition than houses with large living areas.

After excluding these houses, the leverage plots look much better, with little evidence of additional influential points, except for one very small and low-priced house (Supplementary Figure 3).

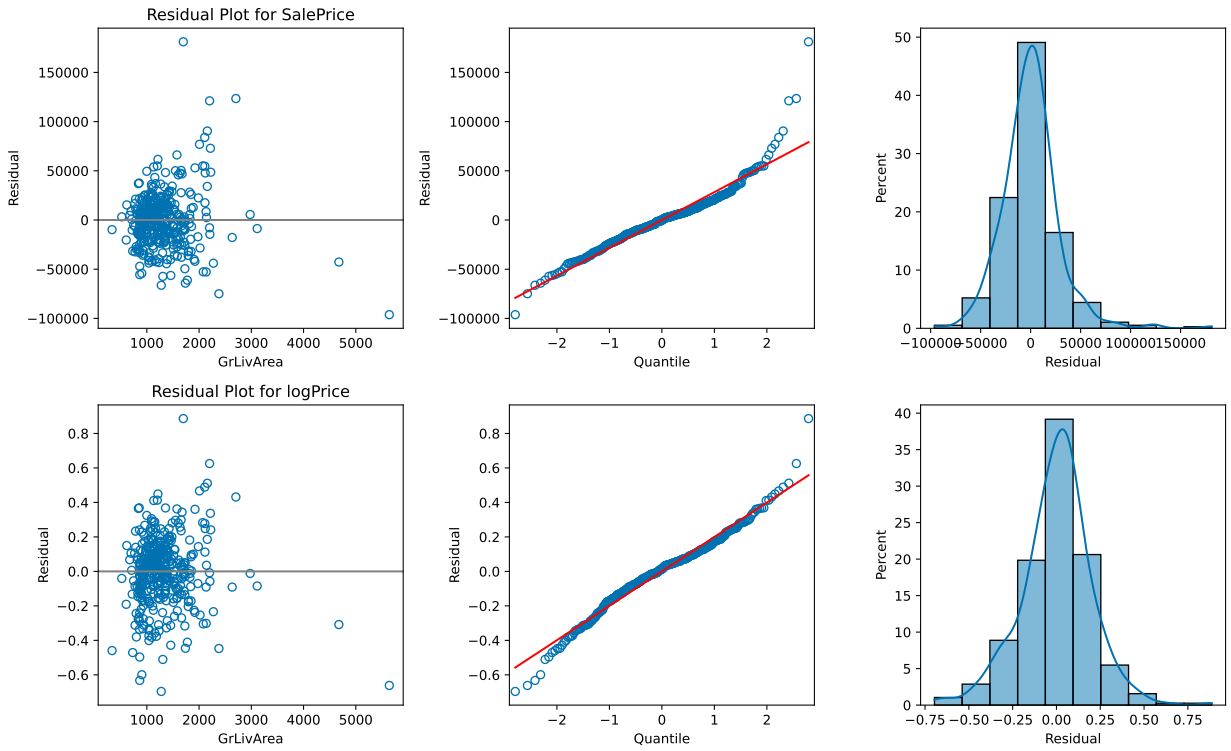


Figure 2: From left to right, residual plots, QQ plots, and histograms of untransformed data (top) and data with log transformation of sale price (bottom), for the model with living area, neighborhood, and their interaction.

Address Assumptions

1. Linearity: Examining the scatterplots of the original and log-transformed price data in Figure 1, the log-lin transformed data appear to provide a better linear fit.
2. Equal Standard Deviation: The residual plot of the untransformed data in Figure 2 shows some evidence of heteroscedasticity. The log-lin transformation reduces this, showing more random scattering with no evidence against equal standard deviation.
3. Normality: Both the original and log-transformed histograms of the residuals in Figure 2 are consistent with normal distributions. The QQ plot of the residuals in the log-lin transformed data is more consistent with a normal distribution than that of the untransformed data.

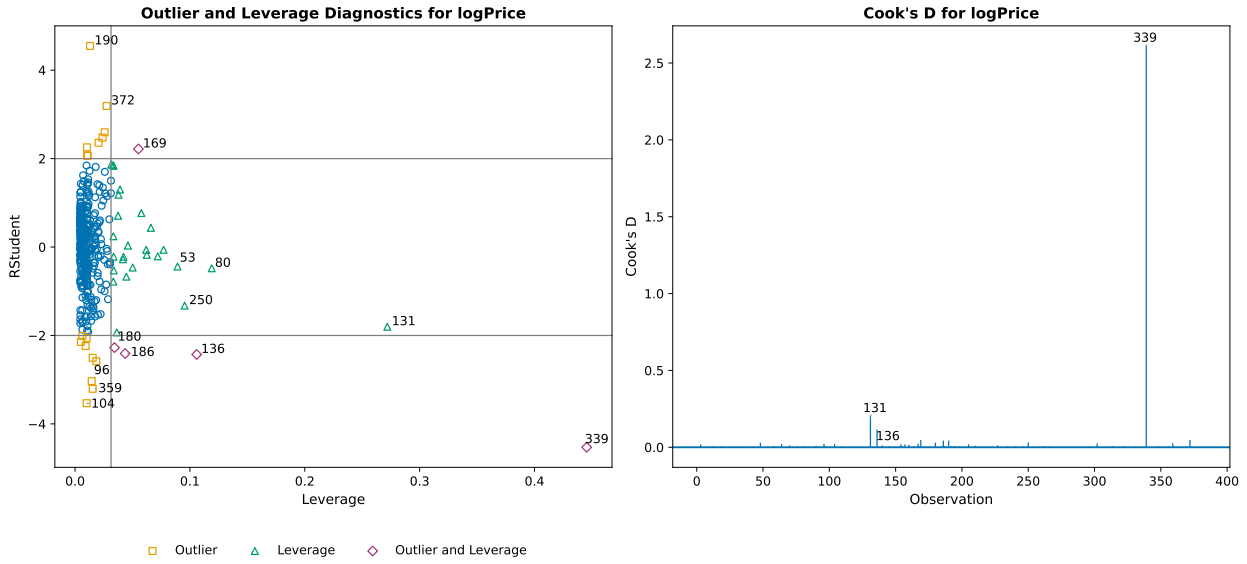


Figure 3: Plots of studentized residuals and leverage (left) and Cook's D (right) for the log-lin model with neighborhood interaction, before removing partially completed houses from the analysis.

4. Independence: I have no evidence against independence. Each observation is a separate home sale, and nothing suggests the sales depend on one another.

Comparing Competing Models: Adj R^2, Internal CV PRESS, AIC

Table 1: Comparison of statistics for the three candidate models, all fit to the data after removing partially completed houses. Higher adjusted R^2 and lower AIC, SBC, and PRESS indicate a better model. The best value in each row is in bold.

Statistic	Log price, independent slopes	Log-log, independent slopes	Log-log, same slope
Adjusted R^2	**0.5263**	**0.5263**	0.5045
AIC	**−877.32**	−877.29	−862.24
SBC (BIC)	**−1234.69**	−1234.67	−1227.49
PRESS	13.846	**13.805**	14.323

All the statistics in Table 1 indicate that models using log transformation with independent slopes for each neighborhood perform the best among the candidate models. This is consistent when comparing additional models not shown, which did not account for the neighborhood or include data transformation. The model with log-transformed price but untransformed area performed about the same as the model with both variables log-transformed. I opted for this model due to its easier interpretation. The AIC and SBC are the lowest for this model and are lower than the models with log-log transformations with or without interaction variables. Its internal CV PRESS is nearly the same as that of the log-log model with independent slopes (13.846 compared with 13.805) and lower than that of the log-log model with the same slope (14.323). The adjusted R^2 is the same for the equivalent model with log-log transformations and is higher than those without interaction variables.

Parameters

Estimates

The coefficient estimates for the general model are in Table 2, and the estimates for each neighborhood, with confidence limits, are in Table 3. The regression line for each neighborhood is shown in Figure 4.

General model:

$$\begin{aligned}\mu\{\log(\text{Price}) \mid \text{Area, Neighborhood}\} = 11.4433 &+ 0.000324\,\text{Area} - 0.6517\,\text{Brookside} \\ &- 0.4334\,\text{Edwards} + 0.000414\,(\text{Area} \times \text{Brookside}) \\ &+ 0.000222\,(\text{Area} \times \text{Edwards})\end{aligned}$$

Neighborhood specific models:

$$\begin{aligned}\mu\{\log(\text{Price}) \mid \text{Area, Neighborhood = North Ames}\} &= 11.4433 + 0.000324\,\text{Area} \\ \mu\{\log(\text{Price}) \mid \text{Area, Neighborhood = Edwards}\} &= 11.0100 + 0.000546\,\text{Area} \\ \mu\{\log(\text{Price}) \mid \text{Area, Neighborhood = Brookside}\} &= 10.7916 + 0.000738\,\text{Area}\end{aligned}$$

Table 2: Parameter estimates for the chosen model, with 95% confidence limits. North Ames is the reference neighborhood, so the Brookside and Edwards terms are differences from North Ames. Confidence limits for the interaction terms are not the limits for the neighborhood slopes (see Table 3).

					95% Confidence Limits	
Parameter	Estimate	Std. Error	t Value	p-value	Lower	Upper
Intercept	11.443341	0.041874	273.28	<0.0001	11.361002	11.525679
GrLivArea	0.000324	0.000030	10.63	<0.0001	0.000264	0.000384
Brookside	−0.651747	0.091719	−7.11	<0.0001	−0.832098	−0.471395
Edwards	−0.433350	0.082403	−5.26	<0.0001	−0.595382	−0.271318
GrLivArea × Brookside	0.000414	0.000071	5.80	<0.0001	0.000274	0.000555
GrLivArea × Edwards	0.000222	0.000062	3.58	0.0004	0.000100	0.000344

Table 3: Slope and intercept estimates for each neighborhood with 95% confidence limits, from refitting the model with each neighborhood as the reference. Slopes are shown as the percent change in mean sale price per 100 square feet of living area. The p-value is for the slope.

		% change per 100 sq ft			Intercept ($)		
Neighborhood	p-value	Estimate	Lower	Upper	Estimate	Lower	Upper
North Ames	<0.0001	3.29	2.68	3.92	93,278	85,905	101,284
Edwards	<0.0001	5.61	4.50	6.74	60,475	52,598	69,532
Brookside	<0.0001	7.66	6.30	9.04	48,610	41,404	57,071

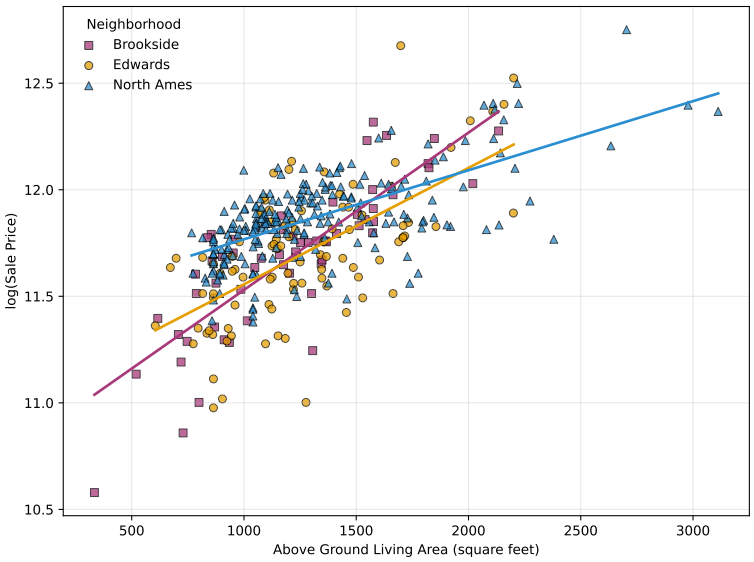


Figure 4: Regression lines of the chosen model, $\mu\{\log(Price) \mid Area, Neighborhood\} = \beta_0 + \beta_1 Area + \beta_2 Brookside + \beta_3 Edwards + \beta_4 Area \times Brookside + \beta_5 Area \times Edwards$*, for each neighborhood. Parameter estimates are in Table 2.*

Interpretation

Slopes: There is convincing evidence at the $\alpha = 0.05$ significance level that for every 100 square foot increase in above-ground living area, the mean expected sale price increases in each of these neighborhoods (Table 3). There is overwhelming evidence that in North Ames, with each additional 100 square feet of living area, the mean expected sale price increases $e^{0.000324\times100} = 1.0329$ times or by 3.29% (p-value < 0.0001). There is overwhelming evidence that in Edwards, the increase is $e^{0.000546\times100} = 1.0561$ times or by 5.61% (p-value < 0.0001). There is overwhelming evidence that in Brookside, the increase is $e^{0.000738\times100} = 1.0766$ times or by 7.66% (p-value < 0.0001).

Intercepts: There is overwhelming evidence that for houses with zero square feet of above-ground living area, the mean expected sale price in North Ames is $e^{11.4433} = \$93{,}278$ (p-value < 0.0001). In Edwards, the mean expected sale price is 35.2% ($1 - e^{-0.4334} = 1 - 0.648$) less than homes in North Ames (Table 2) or $e^{11.0100} = \$60{,}475$ (p-value < 0.0001). In Brookside, the mean expected sale price is 47.9% ($1 - e^{-0.6517} = 1 - 0.521$) less than homes in North Ames or $e^{10.7916} = \$48{,}610$ (p-value < 0.0001). As houses typically have more than zero square feet of living area, this estimate should be interpreted cautiously and in the context of other predictors.

Confidence Intervals

Slopes: Confidence intervals for each neighborhood are in Table 3. I am 95% confident that for every 100 square foot increase in above-ground living area, the mean expected sale price in North Ames increases by between 2.68% and 3.92%, in Edwards by between 4.50% and 6.74%, and in Brookside by between 6.30% and 9.04%.

Intercepts: I am 95% confident that for houses with zero square feet of living area, the mean expected sale price in North Ames is between \$85,905 and \$101,284. In Edwards, it is between \$52,598 and \$69,532 (23.8% to 44.9% lower than North Ames), and in Brookside, it is between

$41,404 and $57,071 (37.6% to 56.5% lower than North Ames). Again, interpreting the sale price for houses with zero square feet of living area should be done cautiously.

Conclusion

In the preceding analysis, I examined the relationship between home sale price and living area within the North Ames, Edwards, and Brookside neighborhoods. The results indicated that this relationship varies by neighborhood. For every 100 square foot increase in above-ground living area, the mean expected sale price increases by 3.29% in North Ames (95% CI: 2.68% to 3.92%), 5.61% in Edwards (4.50% to 6.74%), and 7.66% in Brookside (6.30% to 9.04%). Influential observations were addressed by excluding the four partially completed houses, and model assumptions were verified. A multiple linear regression model with a log transformation of the sale price and interaction terms provided the best fit, enabling generation of estimates and confidence intervals for the relationship between living area and sale price in each neighborhood within the Century 21 Ames market.

Analysis Question 2

Problem

I would like to use linear regression to build a model to predict sale price in all of Ames, Iowa.

Candidate Models

SLR:

$$\mu\{\log(\text{Price}) \mid \text{OverallQual}\} = \beta_0 + \beta_1 \text{OverallQual}$$

MLR 1:

$$\begin{aligned}&\mu\{\log(\text{Price}) \mid \text{OverallQual, } \log(\text{GrLivArea}), \text{ Neighborhood, MSSubClass}\}\\ &= \beta_0 + \beta_1 \text{OverallQual}\\ &\quad + \beta_2 \log(\text{GrLivArea}) + \beta^* \text{Neighborhood} + \beta^* \text{MSSubClass}\end{aligned}$$

where Neighborhood has 25 categories and MSSubClass has 15 categories.

MLR 2:

$$\begin{aligned}&\mu\{\log(\text{Price}) \mid \text{OverallQual, } \log(\text{GrLivArea}), \text{ OverallCond, GarageCars,}\\ &\qquad \text{BsmtFinSF1, YearBuilt, } \log(\text{LotArea}), \text{ Neighborhood}\}\\ &= \beta_0 + \beta_1 \text{OverallQual} + \beta_2 \log(\text{GrLivArea}) + \beta_3 \text{OverallCond}\\ &\quad + \beta_4 \text{GarageCars} + \beta_5 \text{BsmtFinSF1} + \beta_6 \text{YearBuilt}\\ &\quad + \beta_7 \log(\text{LotArea}) + \beta^* \text{Neighborhood}\end{aligned}$$

where Neighborhood has 25 categories.

MLR 3:

$$\begin{aligned}
&\mu\{\log(\text{Price}) \mid \text{OverallQual, } \log(\text{GrLivArea}), \text{ OverallCond, GarageCars,} \\
&\quad \text{BsmtFinSF1, YearBuilt, } \log(\text{LotArea}), \text{ Neighborhood, MSSubClass}\} \\
&= \beta_0 + \beta_1 \text{OverallQual} + \beta_2 \log(\text{GrLivArea}) + \beta_3 \text{OverallCond} \\
&\quad + \beta_4 \text{GarageCars} + \beta_5 \text{BsmtFinSF1} + \beta_6 \text{YearBuilt} \\
&\quad + \beta_7 \log(\text{LotArea}) + \beta^* \text{Neighborhood} + \beta^* \text{MSSubClass}
\end{aligned}$$

where Neighborhood has 25 categories and MSSubClass has 15 categories.

Checking Assumptions

Residual Plots

The residual plots, QQ plots, and histograms of the residuals for the four candidate models are in Figure 5.

Influential point analysis (Cook's D and Leverage)

The studentized residuals and leverage and Cook's D plots for the SLR and MLR 1 models (left half of Figure 6) do not show evidence of any influential points. However, MLR 2 (center right of Figure 6) has points of concern. There was one observation of basement square footage (BsmtFinSF1) above 5000. I decided to restrict the range of the analysis to houses below this threshold. The residuals for MLR 3 benefited from removing this point. The remaining influential point seemed to result from a basement with over 2000 square feet. I considered 4000 as a cutoff, as only two houses fell above 4000. Additionally, only ten houses in the whole dataset were above 2000. However, the house above 4000 and several of the houses above 2000 were in the test set, and I did not want to eliminate them.

Address Assumptions

1. Linearity: Log transformation of SalePrice, GrLivArea, and LotArea gave roughly linear relationships with sale price and more symmetric distributions (Supplementary Figure 4).
2. Equal Standard Deviation: The residual plots in Figure 5 are randomly scattered with no strong evidence against equal standard deviation.
3. Normality: Prior to log transformation, the distributions looked very right-skewed. After log transformation of these variables, the histograms, QQ plots, and random scattering of residuals in Figure 5 are reasonably consistent with a normal distribution. There is some mild left skewness, but I feel comfortable that the sample size will mitigate this.
4. Independence: There is no evidence against independence.

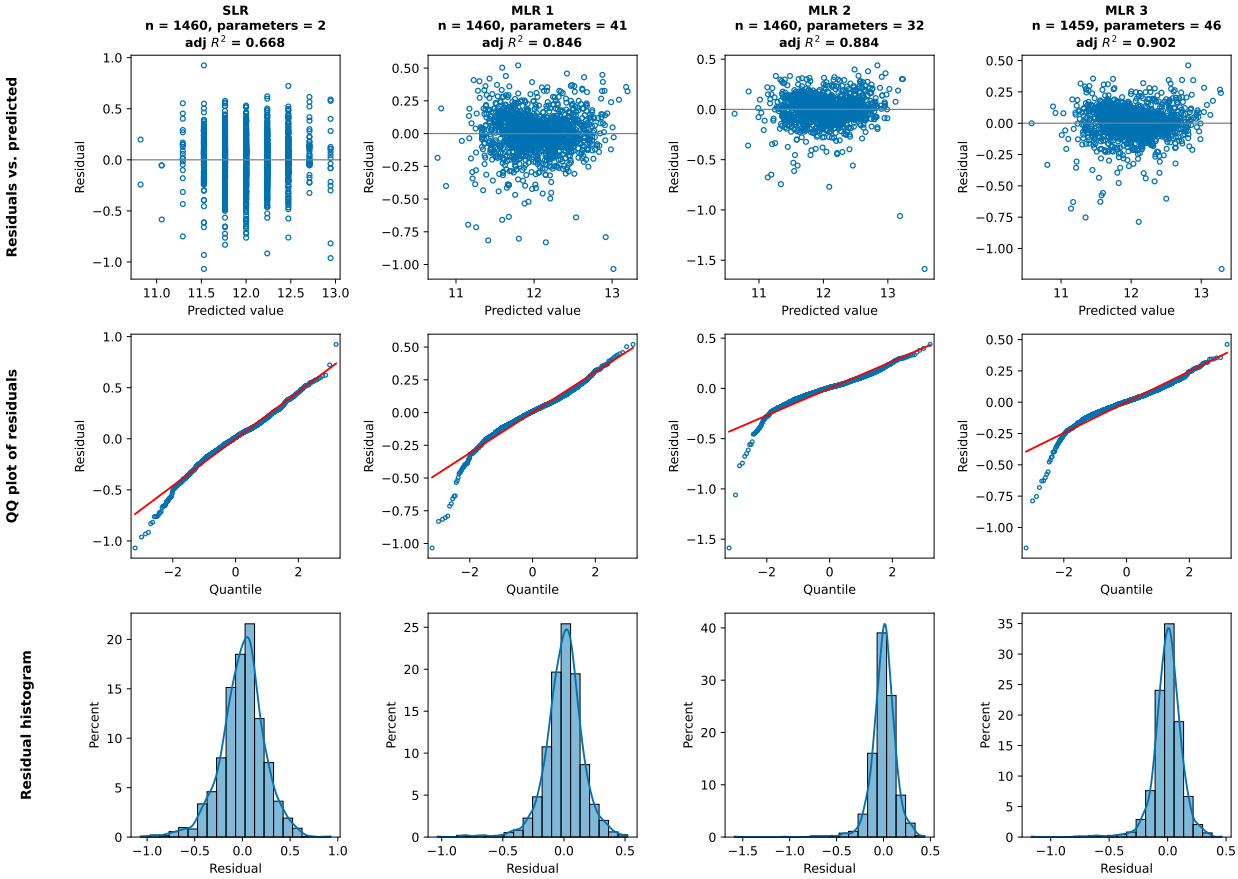


Figure 5: From left to right for candidate models SLR, MLR 1, MLR 2, and MLR 3: residual plots (top), QQ plots (middle), and histograms of the residuals (bottom) after log transformation of selected variables. MLR 3 was fit after restricting the analysis to basement finished square footage under 5000.

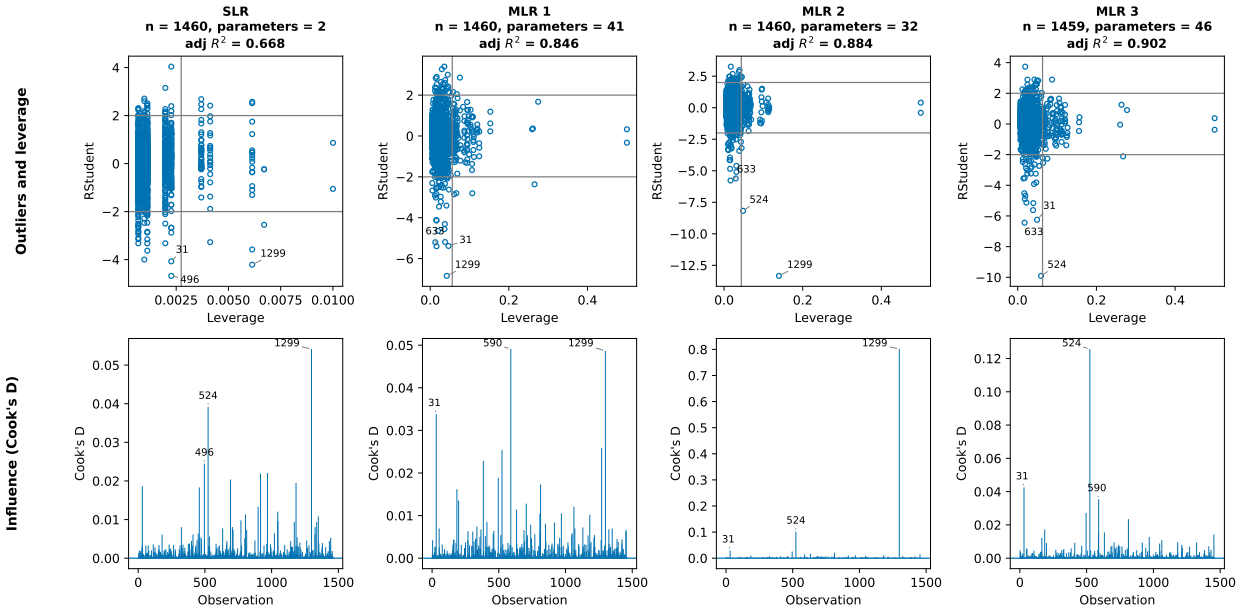


Figure 6: Studentized residuals against leverage (top) and Cook's D (bottom) for candidate models SLR, MLR 1, MLR 2, and MLR 3 (left to right). The three most extreme points in each panel are labeled by observation number. Observation numbers for MLR 3 refer to the dataset after removing the house with basement finished square footage of 5000 or more.

Comparing Competing Models: Adj R^2, Internal CV PRESS, AIC, Kaggle Score

Table 4: Comparison of statistics for the four predictive models, all fit to the 1459 houses with basement finished square footage under 5000. Higher adjusted R^2 and lower PRESS, AIC, and Kaggle score (RMSLE) indicate a better model. The best value in each column is in bold.

Predictive Model	Adjusted R^2	PRESS	AIC	Kaggle Score
SLR	0.672	76.6	−2838	0.229
MLR 1	0.851	36.0	−3952	0.162
MLR 2	0.897	24.7	−4499	0.139
MLR 3	**0.902**	**23.8**	**−4558**	**0.136**

Table 4 shows that MLR 3 has the highest adjusted R^2 and the lowest PRESS, AIC, and Kaggle score. MLR 2 is close behind, and both are much better than the SLR and MLR 1 models.

Conclusion

In Analysis 2, I developed several predictive models for home sale prices in Ames, Iowa. Among the candidate models, the Multiple Linear Regression Model 3 (MLR 3) performed the best, with a high adjusted R^2 of 0.90, and the lowest CV PRESS, AIC, and Kaggle scores. This model included nine predictor variables, two treated categorically and the rest numerically, and employed log transformations of several variables, including the response variable SalePrice. Influential observations and model assumptions were addressed. With one fewer variable (MSSubClass), MLR 2 performed almost as well.

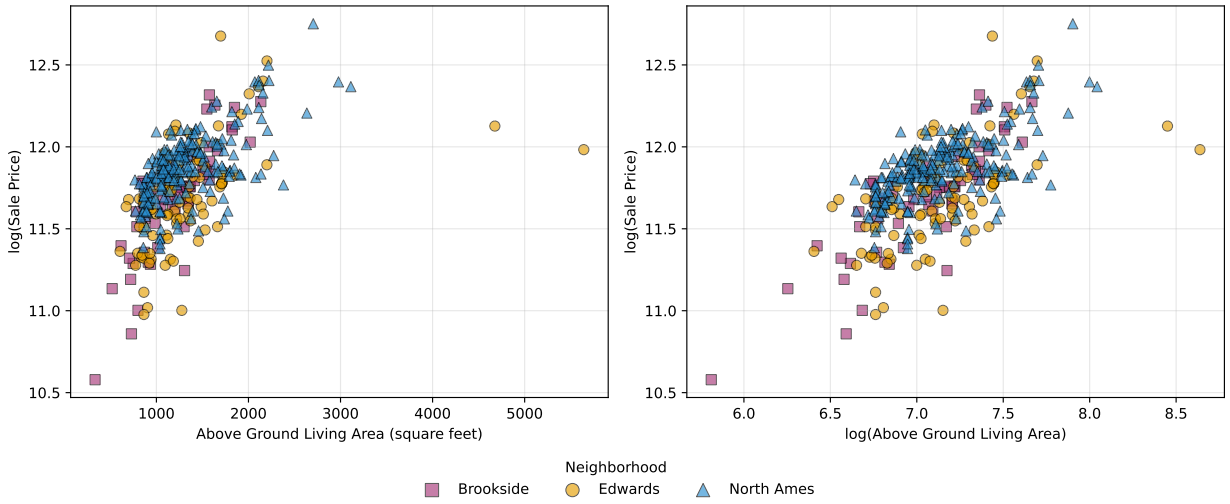
Future improvements could include treating ranked numeric variables as categorical, grouping categories, exploring log transformations on additional variables, and imputing with simpler model predictions rather than median values. Additionally, predicting home prices is highly dependent on the economy and mortgage interest rates, so incorporating these data would benefit future models. MLR 3 performed very well and is the best of these candidate models to predict home sale prices in Ames.

References

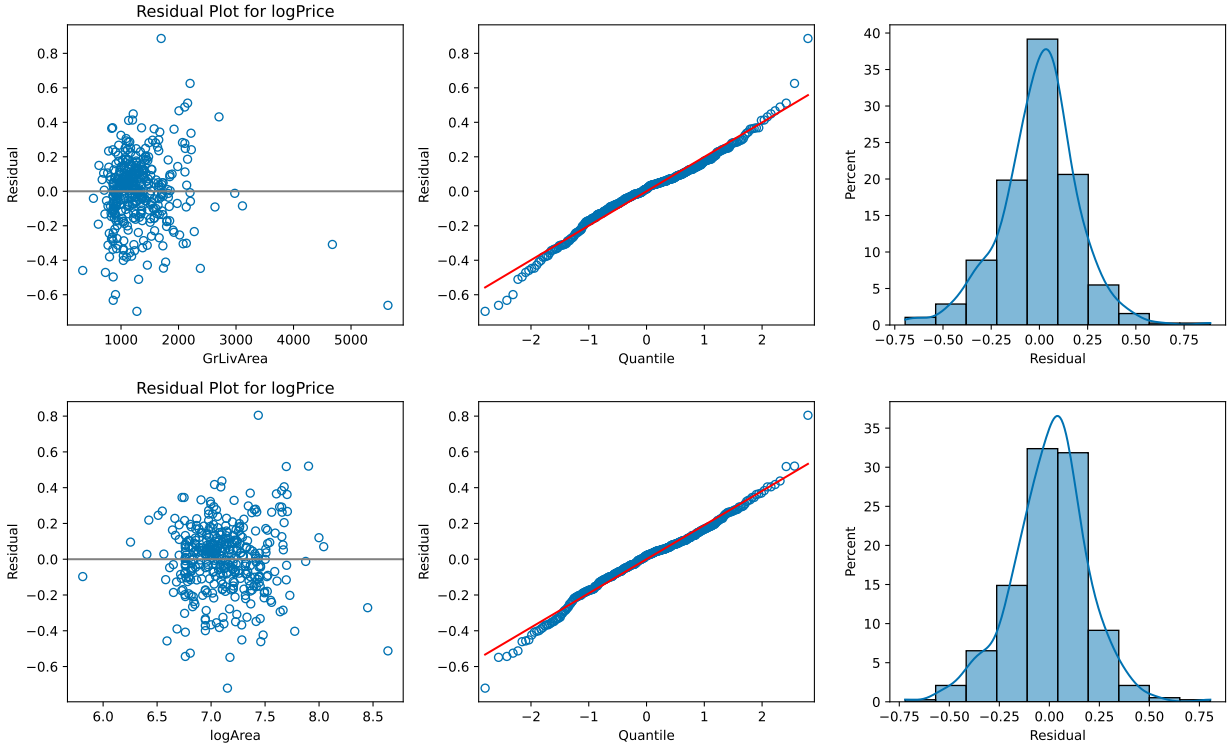
De Cock, D. 2011. "Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project." *Journal of Statistics Education* 19 (3).

Montoya, A., and DataCanary. 2016. *House Prices - Advanced Regression Techniques*. Kaggle. https://kaggle.com/competitions/house-prices-advanced-regression-techniques

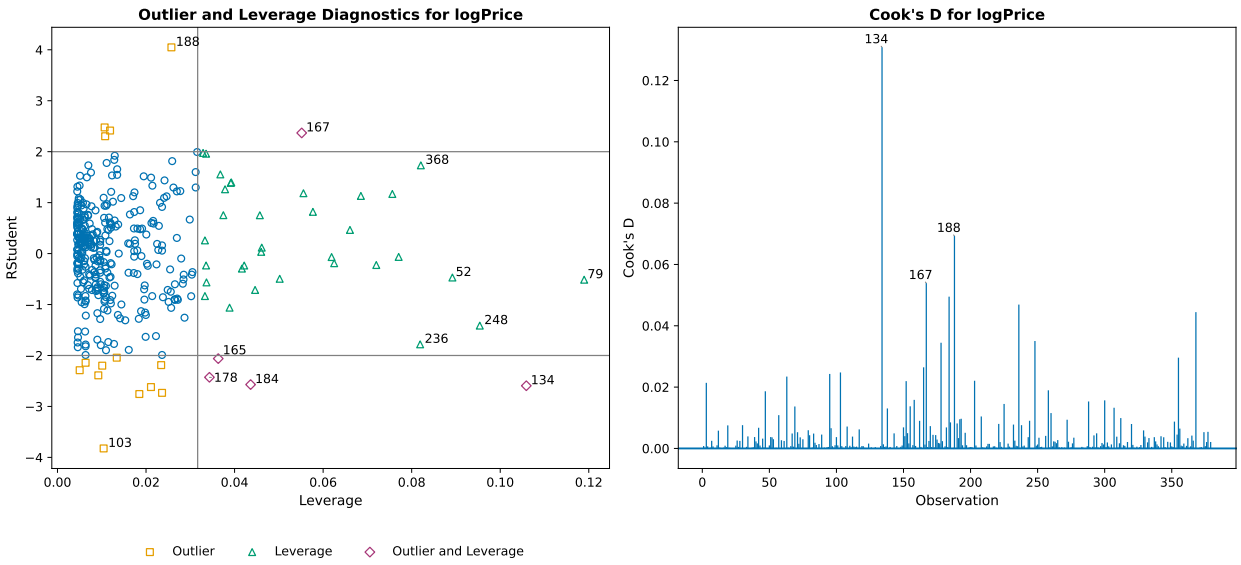
Appendix



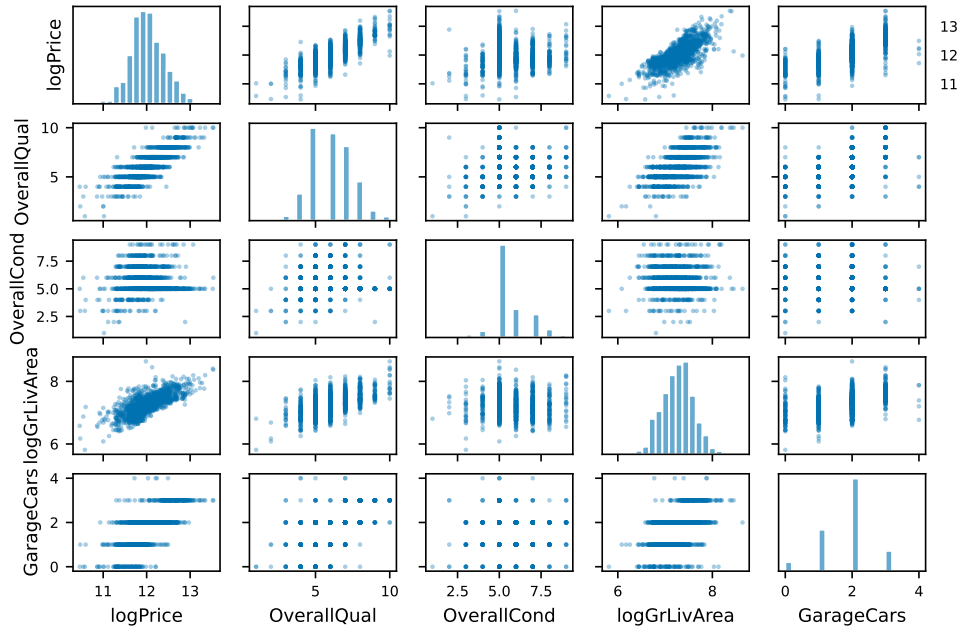
Supplementary Figure 1: Scatterplots of log(sale price) against above-ground living area (left, log-lin) and against log(living area) (right, log-log), with neighborhoods plotted by color and shape (pink-purple squares: Brookside, orange circles: Edwards, blue triangles: North Ames).

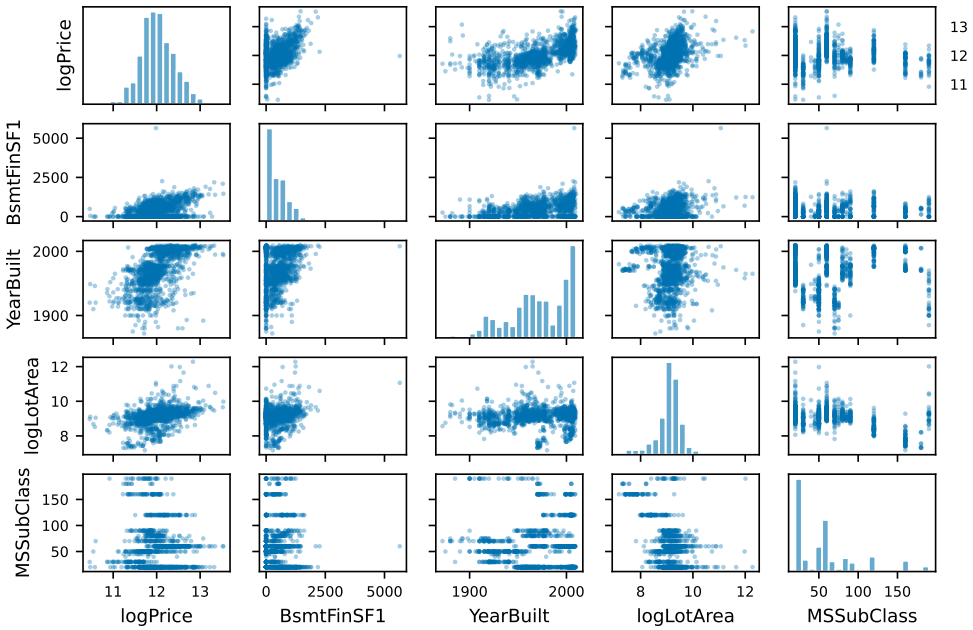


Supplementary Figure 2: From left to right, residual plots, QQ plots, and histograms for the log-lin model (top) and the log-log model (bottom), each with living area, neighborhood, and their interaction.



Supplementary Figure 3: Plots of studentized residuals and leverage (left) and Cook's D (right) for the log-lin model with neighborhood interaction, after removing the four partially completed houses ($n = 379$). Observation numbers refer to the reduced dataset.





Supplementary Figure 4: Scatterplots and histograms of the variables used in the candidate models, shown as two matrices (top and bottom), each including log(sale price). Log transformations were applied to sale price, above-ground living area, and lot area. In each scatterplot the variable named at the left of the row is on the vertical axis and the variable named below the column is on the horizontal axis; the top row of each matrix uses the vertical scale on the right.